§2.1: Resolutions and Smoothings

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1. Abstract

The introduction portion serves as a bit of an "elevator pitch" for the rest of the paper. The general idea of the paper is about the relationship between Stein fillings and Milnor fibres using topological tools. This paper in particular explores the concept of Stein fillings, Milnor fibres, and singularities in an intuitive format. We aim to deduce why the Milnor fibre is worth exploring.

2. Recollection

Recall from the set-up that $(X, 0)$ is some normal surface singularity. We define the resolution $\pi : \tilde{X} \to X$ to be a proper birational morphism such that $\tilde{X}$ is smooth.

- Resolution (§1): simply put, the thing that replaces the singularity. In order to get this "thing," we use a geometric tool known as a "blow-up."
- Blow-up (§1): take a single point (like a singularity) and stick a sphere in its place. The sphere is denoted as $\mathbb{P}^1$.

The first concept of interest is **birational morphisms**.

3. Birational Morphism

Definition 3.1 (Beauville §2.1: Blow-ups)**.** Let S be a surface and $p \in S$. $\exists$ a surface $\hat{S}$ and a morphism $\epsilon : \hat{S} \to S$, which are unique up to isomorphism such that:

i) restriction of ϵ to $\epsilon^{-1}(S - \{p\})$ is an isomorphism onto $S - \{p\}$.

ii) $\epsilon^{-1}(p) = E$, say, is isomorphic to $\mathbb{P}^1$.

By the conditions described above, ϵ is the blow–up of S at point p and E is the exceptional curve. Let p denote out singularity. This is further defined in the Plamenevskaya and Starkson paper. Recall that X is the complex surface with an isolated singularity located at the origin. We let the morphism be: $\pi : \tilde{X} \longrightarrow X$. By condition (i), we are limiting our view to only shapes/structures in question. Namely, this would be $\tilde{X}$ and X. We can further define the surface as: $X = (X - \{p\})$. Similarly, we describe $\tilde{X}$ as: $\tilde{X} = \pi^{-1}(X - \{p\})$. To briefly review the qualification of an isomorphism, we require the following:

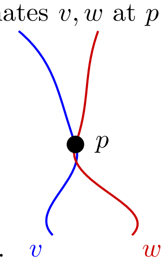
- Bijectivity
- Smooth

Restriction (i) tells us the shape/structure doesn't change at places outside or far from the point p.

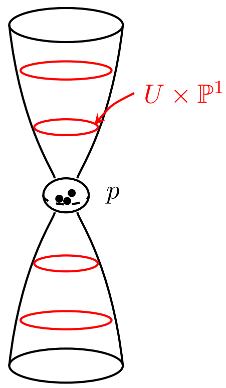
Restriction (ii) describes the shape $\hat{X}$ created by the blow-up at the singularity p. This is E, and as addressed, is isomorphic to $\mathbb{P}^1$.

4. Constructing the Blow-up π

We take a neighborhood $U(p)$. Here, there exists coordinates v, w at p such that $v = w = 0$. Note that the



curves v and w intersect p transversely. To visualize this: We also assume p is the only point of intersection between these curves. The subvariety $\tilde{X}$ of $U \times \mathbb{P}^1$: Note that this is technically a 4D complex space. It just makes life that much easier when you can visualize things. Richard Feynman said: "A picture is worth 1000 words."



is defined by the following equation:

$$0 = uY - vX \ldots (1) \tag{1}$$

$X : Y$ are coordinates in $\mathbb{P}^1$.
From the equation,

$$uY = vX$$
$$\frac{Y}{X} = \frac{v}{u}$$

These are slopes! Let the LHS $= m$
$\therefore v = mX$

From eq. (1), if we look at points very far from the origin, there is at least one pair of $[X : Y]$ that satisfies the equation; bijectivity is satisfied here.
However near the origin, or even at $(x, y) = (0, 0)$, we obtain

$$0 = 0$$

This is true for not none, but all possible choices of $X : Y$. This is our "blow-up." $\therefore$ the pre-image of p is no longer a singularity, expanding into a copy of $\mathbb{P}^1$. Mathematically, we notate this as:

$$\pi^{-1}(p) = \{p\} \times \mathbb{P}^1$$

$\pi^{-1}(p)$ is what we denote as the exceptional divisor.
Our complex surface X is then:

$$U \setminus \{p\} \cong \tilde{X} \setminus \pi^{-1}(p)$$

One problem that we face is that resolutions are not unique. Upon performing a blow-up, we can take a point on the newly created line and perform a blow-up again. This can keep going on forever. The idea of a minimal resolution is to identify the simplest one.

5. Castelnuovo's Contractibility Theorem

Theorem 5.1. *Let X be a complex projective surface and $E \subset X$ a curve isomorphic to $\mathbb{P}^1$, with $E^2 = -1$. There is a morphism $f : X \to Y$ with Y a complex projective surface, such that f is the blow-up of a smooth point $p \in Y$ and E is the exceptional divisor.*

Essentially, this theorem says by blowing up a smooth point on a surface, you make an exceptional curve of the first kind. This curve has the following defining traits:

i) genus $= 0$ (topologically this is a sphere; this we know to be equivalent to $\mathbb{P}^1$).

ii) self-intersection $= -1$; we know this means how the curve twists w/in the surface. The main importance here is that given the value, we are allowed to run the blow-up process in reverse ("blow-down."); essentially, we can collapse the sphere back into a point.

Recall X is our complex surface. Our new shape $\tilde{X} = E$, which was created by the blow-up, is isomorphic to $\mathbb{P}^1$.

We say $E \subset X$ since E is very small in comparison to X. One important thing to note is $E^2 = -1$. Rather than trying to view E as a number, it is viewed as a space.

$$E^2 = -1$$

This means intersection $(E \cdot E)$. It counts how many times E crosses a slightly perturbed version of itself. This happens at exactly one point, which is $= -1$.

Given this, $\tilde{X}$ is our complex surface and $E_v \subset \tilde{X}$ is a curve $\cong$ to $\mathbb{P}^1$. We note that $E_v \cdot E_v = -1$. Then, there is a blow-down morphism $f : \tilde{X} \to X'$ to a simpler smooth surface X', such that f contracts the curve E_v to a smooth point in X'. Here is where $\pi^{-1}(p)$ has normal crossings. This means the curves intersect in the cleanest way possible. We say:

$$\pi^{-1}(p) = \bigcup_{v \in G} E_v,$$

which is the union of the collection of exceptional curves E that can replace the singularity at p (which we denote as a point).

Definition 5.2 (Double-points)**.** A resolution is said to have double-points if only two curves are allowed to intersect at one point. We note that a resolution with this property is a good resolution.

The topology of a good resolution is encoded (or visualized) by the (dual) resolution graph G.

6. Resolutions of Singularities

Essentially, this section aims to show how X' is formed from X. Recall that X is our complex structure with a singularity. X' denotes a complex structure without the singularity.

Definition 6.1 (Resolution)**.** The complex surface X' is a resolution (or desingularization) of the singular complex surface X if X' is a smooth complex surface with a holomorphic map

$$\pi : X' \to X$$

such that π is a biholomorphism away from the singular points $\mathrm{Sing}(X)$. That is, the restriction

$$\pi : X' - \pi^{-1}(\mathrm{Sing}(X)) \to X - \mathrm{Sing}(X)$$

is a biholomorphism.

We note that biholomorphism $\neq$ birational morphism; the paper uses birational morphism. This is more of a global condition, unlike the biholomorphism, which is much stricter.

This book section goes on to note an important property. Normal surfaces only admit isolated singularities. The isolated singularity can be thought of as a branched cover of $\mathbb{C}^2$ branched along a singular curve.

$\hookrightarrow$ we note that a branched cover is a map between spaces that behaves like a standard multi-sheeted covering everywhere except along a specific subset called the branch locus.

What this is telling us is that instead of analyzing the singularity, we can explicitly note it as an algebraic equation branching over a curve. This shifts our attention of a singular curve in $\mathbb{C}^2$.

Recall that some branch curve $B = \{f(x,y) = 0\}$ is resolved by performing repeated blow-ups. Afterwards, the resultant curve B' has normal crossing singularities. Mathematically, this is modeled on:

$$\left\{(x,y) \in \mathbb{C}^2 \;\middle|\; x^l y^m = 0\right\}$$

The blow-up can be described by:

$$\pi : \mathbb{C}^2 \# n\overline{\mathbb{CP}^2} \to \mathbb{C}^2, \quad \text{where}$$

- $\mathbb{C}^2$ – 4D real space written by 2 complex dimensions.
- $\#$ – connected sum operator, which means gluing two manifolds together along a boundary sphere.
- $\overline{\mathbb{CP}^2}$ – the complex projective plane of reversed orientation.
- $n\overline{\mathbb{CP}^2}$ – gluing n separate copies of these orientation-reversed projective planes.

The mathematical expression for π ensures that $\pi^{-1}(p)$ has normal crossings. The important pieces of topological data regarding the exceptional curves (E) is that they possess a self-intersection number of -1. The mathematical expression for π ensures that $\pi^{-1}(p)$ has normal crossings. The important pieces of topological data regarding the exceptional curves (E) is that they possess a self-intersection number of -1.

Again, we note two important results following the blow-up:

i) all curves are smooth and intersect each other in normal crossings

ii) The i^{th} rational curve is decorated by a negative integer e_i (self-intersection) and a positive integer m_i (multiplicity).

Configurations of curves in $\pi^{-1}(0)$ can be described by a dual graph. Each vertex stands for a rational curve, and an edge means the corresponding curves intersect each other.

7. Deformation Theory

The idea behind deforming things is to perturb a given object slightly so that the deformed object is simpler, but carries enough information about the original object.

7.1. Deformation of Complex Space Germs

Note the general definition of a complex space germ. We define it as a concept that isolates the properties of a space infinitely close to a single point, "ignoring everything else that happens further away."

Definition 7.1. Let (X,x) and (S,s) be complex space germs. A deformation of (X,x) over (S,s) consists of a flat morphism $\phi : (\mathcal{X},x) \to (S,s)$ of complex germs together with an isomorphism from (X,x) to the fibre of ϕ,

$$(X,x) \to (\mathcal{X}_s, x) := (\phi^{-1}(s), x).$$

$(\mathcal{X},x)$ is called the total space, (S,s) the base space, and $(\mathcal{X}_s,x) \cong (X,x)$ the special fibre of the deformation.

In relation to this paper, the surface singularity is $(X,0)$. Its deformation is the flat mapping:

$$\lambda : (\mathcal{X},0) \to (T,0) \quad \text{where} \quad \lambda^{-1}(0) = (X,0).$$

This is saying that at the origin, the fibre is isomorphic to the original shape X. This is our special fibre.

Away from the origin, where $\lambda^{-1}(s)$ with $s \neq 0$, the fibre yields a new smoothed out shape $\tilde{X}$. We call this the general fibre for now.

7.2. Versal Deformations

A versal deformation of a complex space germ is a deformation which contains basically all information about any possible deformation of this germ. A fundamental fact is that any isolated singularity has a versal deformation.

Any other deformation (X, x) over a base space (T, t) can be induced by (i, ϕ), the original deformation by a base change:

$$\varphi : (T, t) \to (S, s)$$

If a deformation of (X, x) over a subgerm $(T', t) \subset (T, t)$ is given and induced by a base change:

$$\varphi' : (T', t) \to (S, s),$$

we can choose φ in a way that it extends to φ'. Essentially, this gives us a way to add space to a versal deformation. All that's really required is that in the paper regarding versal deformations:

A deformation $\lambda : (\mathcal{X}, 0) \to (D, 0)$ over the disk in $\mathbb{C}$ is called a (1-parameter) smoothing of $(X, 0)$ if $X_s := \lambda^{-1}(s)$ is smooth $\forall\ s \neq 0$. This means in order for the definition of the smoothing to hold, then all points far from the point $s = 0$ must be smooth. This yields our new shape $\tilde{X} = X_s$, to which we call our Milnor fibre.

Recall that Milnor fibres are a type of Stein structure. In this case, it is that for parameters in the disk, excluding the origin, $X_{s_0} = X$ and $X_{s_1} = \tilde{X}$ are Stein homotopic.

A Stein homotopy means that we can take X_{s_0}, deform it continuously, and obtain X_{s_1}. Both shapes will be diffeomorphic and contact structures remain equivalent throughout their deformation.

8. The Compact Version of the Milnor Fibre to Obtain a Stein Filling of the Link Y of $(X, 0)$

Unsmoothed singularities extend to infinity. In order to understand them topologically, we cut them down to a compact 4-manifold with a boundary. The goal is to establish a domain that naturally inherits a Stein structure to fill Y (a 3-manifold).

Since the singularity is at the origin, choosing $r > 0$ where $X \subset \mathbb{C}^N$ is transverse to a sphere S_r^{2N-1} ensures that the intersection is smooth and well-behaved within the manifold.

Fixation of a ball $B_r^{2N} \subset \mathbb{C}^N$ centered at 0 (Milnor Ball) is not necessarily a germ, but a neighborhood. When $X \cap B_r^{2N}$ (intersection), we gain all information about the singularity point.

We note that $\partial(X \cap B_r^{2N})$ is the link Y of $(X, 0)$. Recall that Y is the 3D manifold created by slicing the complex surface with the boundary of the ball.

Let X be the complex surface passing through the origin. B_r^{2N} is a closed ball of radius r, making its boundary the sphere S_r^{2N-1}.

$$\therefore \partial B_r^{2N} = S_r^{2N-1}$$

To slice the boundary,

$$\partial(X \cap B_r^{2N}) = X \cap \partial B_r^{2N} = X \cap S_r^{2N-1}$$

Recall that the complex tangencies on the surface X intersecting the boundary are the 2D contact structure, denoted as ξ.

We know that where $s \neq 0$, the structure is smooth in X. Depending on how small s is, X_s inherits the same intersection of the sphere boundary that the original singularity had.

$X_s \cap B_r^{2N}$ is diffeomorphic to Y means the boundary of the smoothed fibre matches the boundary of the original structure. $X_s \cap B_r^{2N}$ is a complex submanifold in $\mathbb{C}^N$ (has properties of Stein filling).

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